

Machine Learning Algorithms Laboratory

Experiment 8 Report

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering 1
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Submission Date	

1 Aim and Objective

The aim of this experiment is to implement and compare three clustering algorithms, K-Means, DBSCAN and Hierarchical Agglomerative Clustering (HAC), on the Human Activity Recognition (HAR) dataset. The clusters formed by each algorithm are visualized and evaluated using internal metrics (which do not need true labels) and external metrics (which compare clusters against the actual activity labels), even though clustering itself is an unsupervised technique.

2 Dataset Description

Dataset	Description
Dataset Name	Human Activity Recognition Using Smartphones Dataset
Number of Samples	10299 (7352 train + 2947 test, combined)
Number of Features	561 (time and frequency domain features from accelerometer/gyroscope signals)
Activity Labels (for evaluation only)	WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING
Missing Values	None
Duplicate Rows	None

3 Preprocessing Steps

The training and test partitions provided in the dataset were combined into a single dataset of 10299 samples, since clustering is unsupervised and does not need a train-test split in the usual sense. There were no missing values or duplicate rows. All 561 features were standardized using `StandardScaler` since K-Means and DBSCAN are both distance based and are sensitive to feature scale. For visualization purposes, PCA was applied to reduce the 561 features down to 2 principal components.

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

print("Explained variance:", pca.explained_variance_ratio_.sum())
```

The 2 principal components together explain only about 57% of the total variance, so the 2D scatter plots used for visualization in this report should be seen as a simplified view of the actual 561-dimensional clustering, not the full picture.

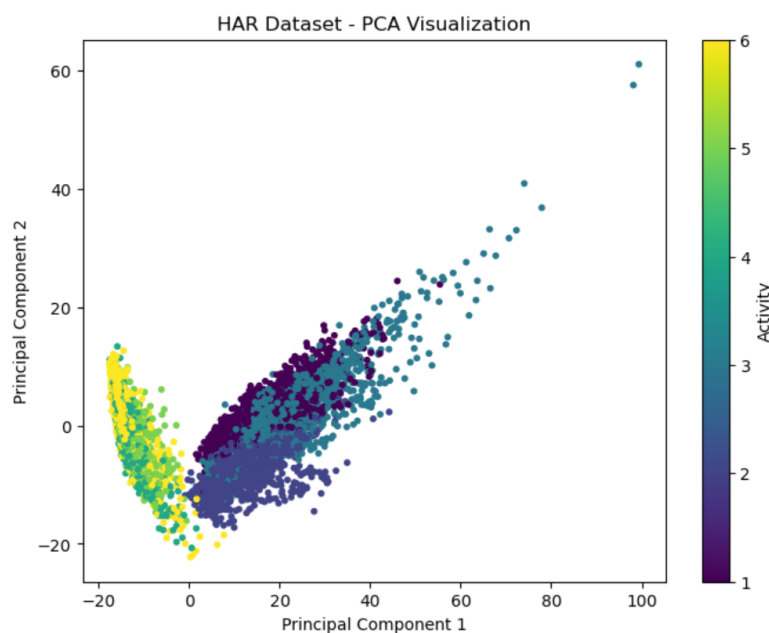


Figure 1: 2D PCA Projection Colored by True Activity Label

Inference: Even in just 2 principal components, the data roughly separates into two broad groups, the "static" activities (Sitting, Standing, Laying) form one cluster and the "moving" activities (Walking, Walking Upstairs, Walking Downstairs) form another, with the three static activities overlapping quite a lot with each other. This gives an early hint that unsupervised clustering may find 2 natural groups rather than the full 6 activities.

4 K-Means Clustering with Elbow Method

K-Means was run for $k = 2$ to 8, recording the WCSS (inertia) and Silhouette Score for each value of k .

```
for k in k_values:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_scaled)
    wcss.append(km.inertia_)
```

```
silhouette_scores.append(silhouette_score(X_scaled, km.labels_))
```

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	3272857	0.3937
3	2921078	0.3155
4	2781602	0.1500
5	2654790	0.1286
6	2577180	0.1099
7	2519774	0.0816
8	2465379	0.0749

Table 2: K-Means Elbow Method Results

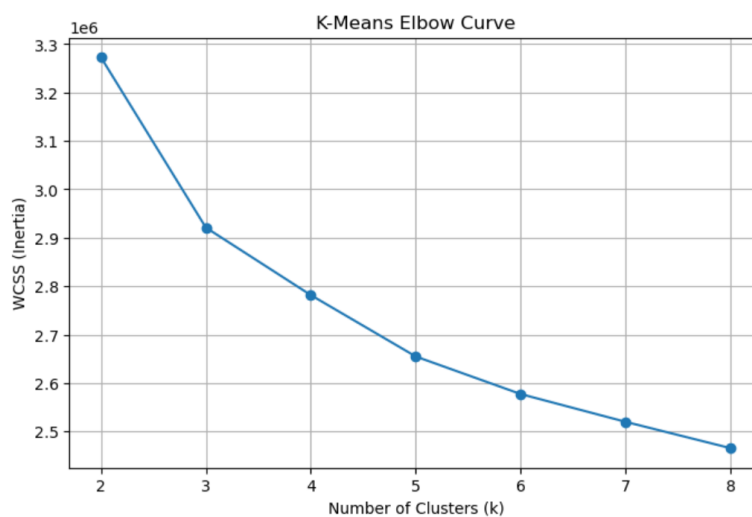


Figure 2: Elbow Curve - k vs WCSS

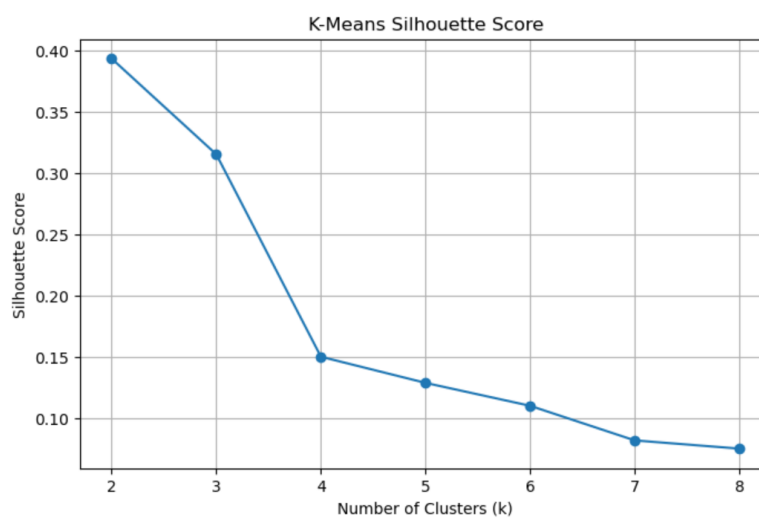


Figure 3: k vs Silhouette Score

Inference: The WCSS curve drops sharply going from $k=2$ to $k=3$ and then flattens out gradually, without a very sharp "elbow", making the elbow point a bit ambiguous to read visually. The Silhouette Score, however, is clearly highest at $k=2$ (0.3937) and drops noticeably for every k beyond that. Since silhouette score gives a clearer and more quantitative signal here, $k=2$ was chosen as the best value, which also lines up with the earlier observation that the data broadly separates into a "static vs moving" grouping rather than 6 separate activities.

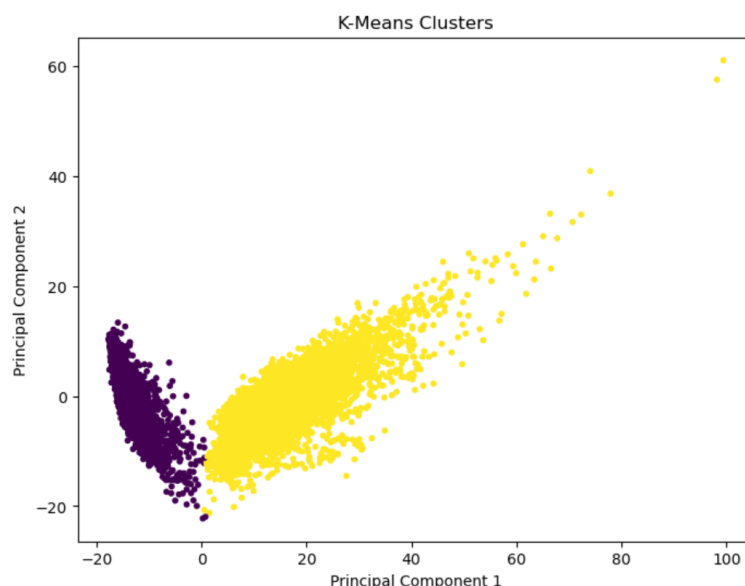


Figure 4: K-Means Clusters ($k=2$) on PCA-Reduced Data

Inference: The two K-Means clusters line up reasonably well with the "static vs moving" split seen in the true label plot, but K-Means being a hard partitioning algorithm cannot separate all 6 individual activities well since it always looks for roughly spherical, equally sized clusters, and the actual activity groups do not really have that shape in the reduced 2D space.

5 DBSCAN Clustering

DBSCAN requires two parameters, `eps` (neighborhood radius) and `min_samples`. Several values of `eps` were tried (with `min_samples` fixed at 5) to see their effect on the number of clusters and noise points found.

```
eps_values = [5, 10, 15, 20, 25]

for eps in eps_values:
    db = DBSCAN(eps=eps, min_samples=5)
    labels = db.fit_predict(X_scaled)
    n_clusters = len(set(labels)) - (1 if -1 in labels else 0)
    n_noise = list(labels).count(-1)
```

Epsilon	Number of Clusters	Noise Points
5	0	10299
10	25	7545
15	19	2029
20	1	434
25	2	127

Table 3: DBSCAN Epsilon Tuning Results

Inference: A very small eps (5) treats every single point as noise since no point has 5 neighbors within that tiny radius in 561-dimensional space, this is a common issue with DBSCAN on high dimensional data. As eps increases, the number of noise points drops sharply, but eps=10 produces an unreasonably large number of clusters (25), most of which are probably tiny and not meaningful. eps=25 gives a much more sensible result, 2 clusters with only 127 points marked as noise, so eps=25 was chosen as the final parameter instead of just picking whichever eps happened to have the "best" looking numbers on paper.

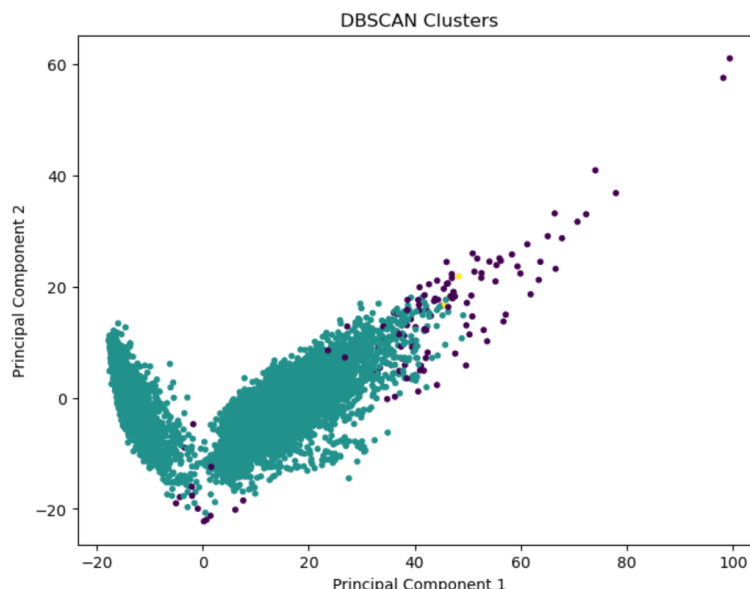


Figure 5: DBSCAN Clusters (eps=25, min_samples=5) on PCA-Reduced Data

Inference: DBSCAN also converges to essentially 2 main clusters (plus a small amount of noise), similar to K-Means, again suggesting that the data naturally has 2 dominant density regions rather than 6.

6 Hierarchical Agglomerative Clustering

Since hierarchical clustering has a high memory/computation cost for large datasets, it was run on a random subsample of 3000 points instead of the full 10299 samples. Ward's linkage was used, which tends to produce more balanced, compact clusters.

```

hac_indices = np.random.choice(len(X_scaled), size=3000, replace=
    False)

```

```

X_hac = X_scaled[hac_indices]

hierarchical = AgglomerativeClustering(n_clusters=2, linkage="
ward")
hierarchical_labels = hierarchical.fit_predict(X_hac)

```

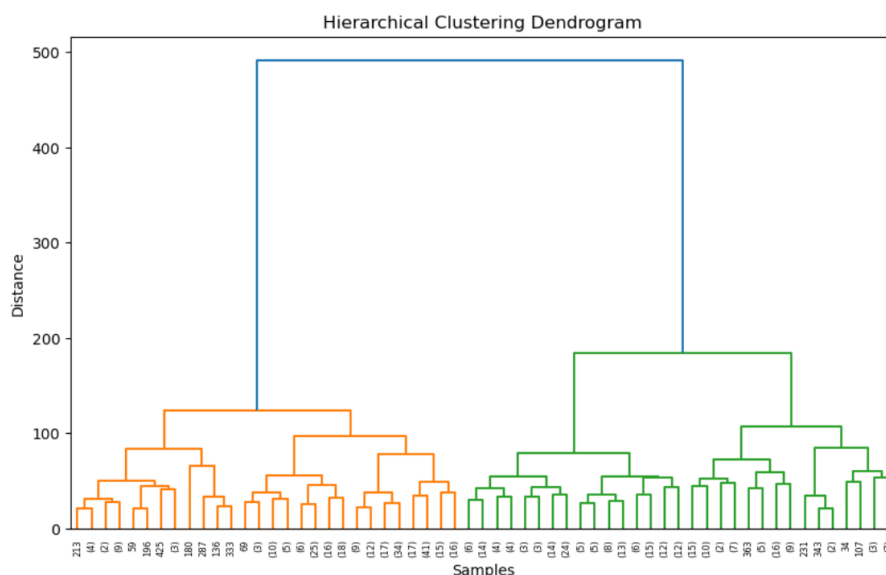


Figure 6: Dendrogram (first 500 samples, Ward Linkage)

Inference: The dendrogram shows a clear, large merge happening quite high up, which visually confirms that cutting the tree into 2 clusters is a natural choice, matching the number of clusters chosen for K-Means and DBSCAN as well.

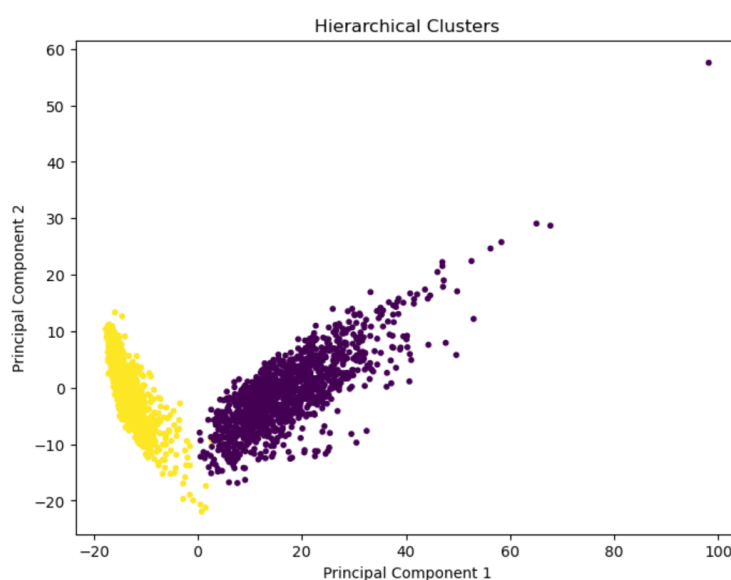


Figure 7: Hierarchical Clusters (n=2, Ward Linkage) on PCA-Reduced Data

Inference: The hierarchical clustering result looks visually very similar to the K-Means result, both algorithms converge on a similar 2-way split, which makes sense since

Ward's linkage also tries to minimize within-cluster variance, similar in spirit to what K-Means optimizes.

7 Evaluation Metrics and Results

Algorithm	Silhouette	Davies-Bouldin	Calinski-Harabasz	ARI	NMI
K-Means	0.3937	1.0707	7880.81	0.3296	0.5455
DBSCAN	0.4841	0.6337	18.87	0.0016	0.0142
Hierarchical	0.3954	1.0685	2312.16	0.3323	0.5548

Table 4: Internal and External Clustering Evaluation Metrics

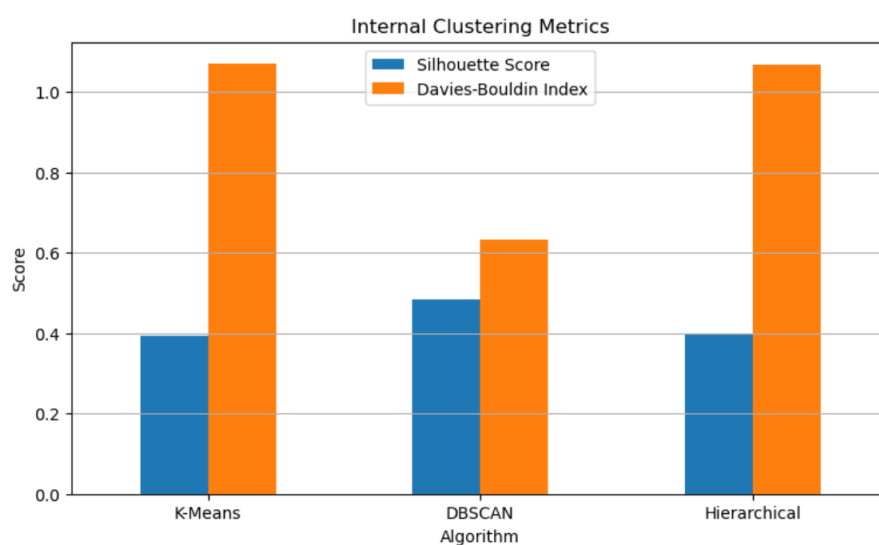


Figure 8: Internal Clustering Metrics Comparison

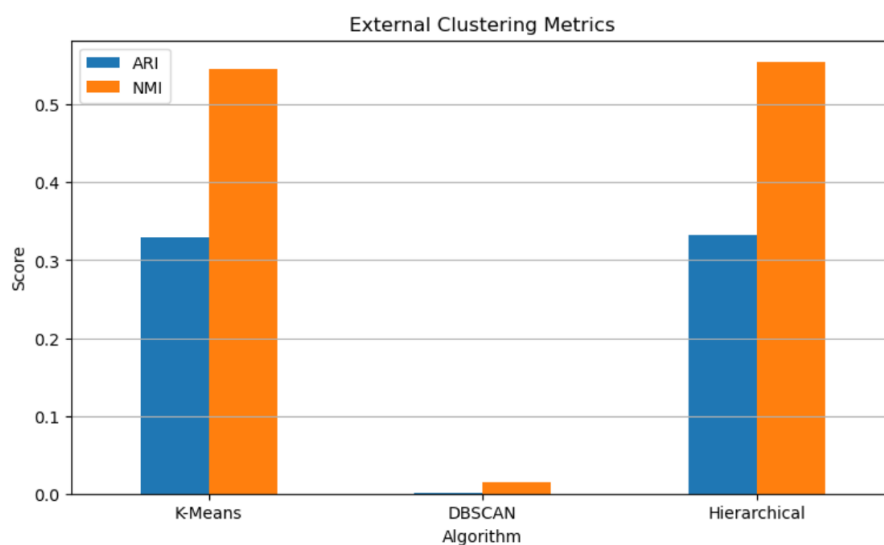


Figure 9: External Clustering Metrics Comparison (ARI, NMI)

Inference: This is one of the more interesting results in the experiment. Based on internal metrics alone, DBSCAN actually looks the best, it has the highest Silhouette Score (0.4841) and the lowest (best) Davies-Bouldin Index (0.6337). But its Calinski-Harabasz Index (18.87) is drastically lower than the other two, and its external metrics (ARI=0.0016, NMI=0.0142) are almost zero, meaning its clusters barely correspond to the real activity labels at all. K-Means and Hierarchical Clustering, on the other hand, have much higher and near-identical ARI/NMI values (around 0.33 and 0.55), meaning their 2-cluster split actually lines up reasonably well with the true "static vs moving" grouping, even though clustering never saw the labels during training. This shows that no single metric tells the whole story, and internal metrics alone (which only look at the shape/compactness of clusters) can strongly disagree with how well the clusters match a meaningful real-world grouping.

8 Observation Questions

1. Which algorithm produced the most meaningful clusters? Why?

Based on how well the clusters match the true activity labels (ARI and NMI), Hierarchical Clustering and K-Means produced the more meaningful clusters (ARI around 0.33, NMI around 0.55 for both), since their 2-cluster split closely follows the natural "static vs moving" activity grouping. DBSCAN, despite scoring the best on purely internal shape-based metrics, produced clusters that barely relate to the true activity labels (ARI ≈ 0), so in terms of real-world meaning, it was the least useful here.

2. How sensitive was K-Means to the choice of k ?

Quite sensitive. The Silhouette Score dropped from 0.3937 at $k=2$ to 0.3155 at $k=3$ and then kept falling all the way down to 0.0749 at $k=8$. This shows that K-Means found a strong, well-separated 2-cluster structure in this data, and forcing it to find more clusters (closer to the actual 6 activities) only produced weaker, less well separated partitions.

3. Did DBSCAN detect noise or small clusters effectively?

DBSCAN was extremely sensitive to the ϵ parameter, going from labelling 100% of points as noise at $\epsilon=5$, to 25 fragmented clusters at $\epsilon=10$, and eventually stabilizing at just 2 clusters with about 1.2% noise points at $\epsilon=25$. This shows that DBSCAN can identify noise effectively, but only once a suitable ϵ is found through trial and error, its behaviour changes drastically with even small changes in ϵ , which is a known challenge with DBSCAN especially in high dimensional spaces like this 561-feature dataset.

4. How does linkage choice (single/complete/ward) affect hierarchical clustering?

Only Ward's linkage was used in this experiment, which tends to produce compact, roughly equal-sized clusters by minimizing the increase in within-cluster variance at each merge. Single linkage, in comparison, tends to produce elongated, chain-like clusters and is more sensitive to noise/outliers (since it merges based on the single closest pair of points), while complete linkage tends to produce more compact but sometimes uneven clusters based on the farthest pair of points. Since the HAR data does not appear to have chain-like structure, Ward's linkage was a reasonable choice and gave a clean, balanced 2-cluster split as seen in the dendrogram.

5. Which internal metric best matched your visual intuition of cluster quality?

The Silhouette Score matched the visual intuition best. Looking at the PCA scatter

plots, K-Means and Hierarchical clustering visually separate the data into two reasonably distinct, non-overlapping blobs, and both had a healthy Silhouette Score around 0.39. The Calinski-Harabasz Index was less intuitive to interpret directly since its scale varies hugely between algorithms (7880 for K-Means vs just 18.87 for DBSCAN), making it harder to judge at a glance compared to Silhouette Score, which is conveniently bounded between -1 and 1.

9 Conclusion

In this experiment, K-Means, DBSCAN and Hierarchical Agglomerative Clustering were applied to the Human Activity Recognition dataset. All three algorithms converged on a similar structure, finding 2 dominant clusters corresponding broadly to "static" and "moving" activities rather than the full 6 individual activities. K-Means and Hierarchical Clustering gave very similar and reasonably meaningful results, both matching the true activity labels moderately well ($\text{ARI} \approx 0.33$). DBSCAN needed careful tuning of ϵ to avoid labelling everything as noise, and even after tuning, its clusters had the best internal compactness scores but almost no correspondence to the actual activity labels. This experiment showed that internal clustering metrics and how well clusters match real-world ground truth labels can tell very different stories, and both should be considered when judging the quality of a clustering result.

10 References

1. Scikit-learn Documentation: Clustering. <https://scikit-learn.org/stable/modules/clustering.html>
2. Human Activity Recognition Using Smartphones Dataset, UCI Machine Learning Repository.